

RÚBEN ARAÚJO

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 Portuguese (EU)

RESEARCH EXPERIENCE (Biomedical / Clinical)

Researcher – Instituto Superior de Engenharia de Lisboa (ISEL) Oct 2017 – Present | Lisbon, Portugal

- Led and supported biomedical and clinical research activities focused on establishing and optimizing laboratory protocols for biospecimen handling, sample preparation, FTIR metabolomics, and proteomics assays across multidisciplinary projects spanning transplantation rejection, infection, mortality, COVID-19, embryo viability, genotoxicity, and food biocompounds. Supervision of MSc students and support to PhD colleagues.

Studentship – Fundação para a Ciência e a Tecnologia (FCT) Nov 2021 – Nov 2025 | Lisbon, Portugal

- Research in ICU infection diagnosis and mortality prognosis using FTIRS serum metabolomics and targeted proteomics (multiplex cytokine assays) in critically ill COVID-19 patients, including delirium classification; infection FTIRS work resulted in a provisional patent filing, with a planned hardware/software commercialization pathway.

Researcher – Comprehensive Health Research Center (CHRC) Nov 2021 – Present | Lisbon, Portugal

- Integrated in multidisciplinary clinical/biomedical research groups; contributed to biomarker-focused studies and collaborative scientific activities.

Principal Investigator – Instituto Português da Face (IPF) 2020 | Lisbon, Portugal

- Led research on TMJ tissue regeneration and 3D surgical planning, including sample/data processing, statistical analysis, and scientific publication.

PUBLICATIONS (2020-2025)

- Peer-reviewed manuscripts: 30 total (first author: 9; co-authored: 21).
- Quartiles: 13 Q1, 13 Q2, 2 Q3, 1 Q4.
- Topics: Infectious diseases, critical care mortality, COVID-19, nephrology, transplantation rejection, genotoxicity, extracellular vesicles, embryo viability, bioactive food compounds.
- Patent: PT120761 (2025): Infection Diagnosis (INPI, Portugal).
- Complete list with DOIs and PDFs available for download at: rubenaraujo.org/cv

RESEARCH PROJECTS

- Participation in 8 funded research projects as: Researcher, Principal Researcher, Founder.
- Combined funding across projects: ≈1 million Euros.
- Domains: clinical metabolomics, biomedical devices, infection diagnostics, critical care, TMJ regenerative solutions, nephrology, machine learning applied to clinical medicine.
- Institutions involved: NOVA Medical School (NMS), ISEL, Centro Hospitalar Universitário de Lisboa (CHULC), Instituto Superior Técnico (IST), Instituto Português do Sangue e da Transplantação (IPST), Instituto Politécnico de Lisboa (IPL).

MEDICAL RESEARCH INNOVATION

- Patent application: PT120761 — Infection Diagnosis (INPI Portugal, 2025).
- Short description: Machine learning and serum metabolomics workflow for rapid infection detection and stratification (presence, bacterial versus fungal, site, gram type, species).
- Biomedical hardware/software concept development for sample processing and data acquisition workflows, towards medicine frameworks enabling trajectory-based risk alerts.

ENGINEERING R&D HIGHLIGHTS

- R&D experience in industrial workflows, rapid prototyping, and systems integration.
- Field-oriented problem solving in multidisciplinary engineering teams (mechanical, hydraulics, control systems, documentation, sales).

EDUCATION

PhD in Biomedicine – NOVA Medical School (NMS) 2021 – 2026 | Lisbon, Portugal

MSc in Biomedical Engineering – ISEL 2017 – 2019 | Lisbon, Portugal

MSc in Mechanical Engineering (Maintenance & Production) – ISEL 2007 – 2011 | Lisbon, Portugal

BSc in Mechanical Engineering – ISEL 2002 – 2007 | Lisbon, Portugal

BIOMEDICAL & CS SKILLS

Sample processing	Protocol handling
Metabolomics	Proteomics
COVID-19	Biomarkers
Intensive Care	Immunology
Infection	Critical Care
Data Analysis	Machine Learning
Python / R	SQL

LANGUAGES

- Portuguese – Native
- English – C1 (IELTS + Cambridge)
- Norwegian – B1 (Aftenskolen)
- Spanish – B1 (Instituto Cervantes)
- French – Basic

ACADEMIC LEADERSHIP

- Erasmus Student Coordinator (ISEL) |2018-2019
- Founder, EMBS Student Branch (ISEL) |2018
- Chair, IEEE Student Branch (ISEL) |2017-2018
- Class Representative Bio. Eng. (ISEL) |2017-2019

AWARDS & SCHOLARSHIPS

- FCT PhD Studentship |2021-2025
- First in Class (PhD) |2023
- Research Scholarship (ISEL) |2021
- Merit Scholarship (Ministry of Education) |2021
- First in Class (MSc Bio. Eng) |2020

ENGINEERING EXPERIENCE

- CEO – Memory Particle (Portugal) |2017-2018
- Lead Mechanical – Brøvig (Norway) |2016-2017
- Life Cycle Engineer – MH Wirth (Norway) |2014-2015
- Project Engineer – Aker Solutions (Norway) |2012-2014
- Mechanical Design – Siemens (Portugal) |2011-2012

Note: Senior engineering experience in large-scale, multidisciplinary and multinational industrial projects, involving coordination across diverse technical teams, communication with multiple stakeholder groups, and interface with executive-level decision makers.

All supporting documentation can be consulted and downloaded @ <https://rubenaraujo.org/cv/>